Deepak Mallampati

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PROFESSIONAL SUMMARY

Applied AI engineer (MS CS, 2.5+ years) leading end-to-end LLM deployments in regulated clinical and life sciences environments. Designed production AI systems with full data provenance, reliability engineering, and compliance-grade audit trails across discovery, clinical, and operational workflows. Delivered ~35% retrieval precision gain, 15-20% improvement in high-risk clinical category recall (precision >90%), and 98% dataset reliability in HIPAA-conscious environments. Expert in hypothesis-driven problem framing, scoping complex enterprise AI deployments, and translating ambiguous life sciences workflow needs into production systems. Open to relocation; open to European work authorization discussions.

CORE COMPETENCIES

End-to-End LLM Deployment (Life Sciences) • Clinical Data AI Systems • RAG & Knowledge Retrieval • Data Provenance & Reliability Engineering • Agentic Workflows • Predictive ML (Clinical Outcomes) • Regulated AI (HIPAA / Compliance) • Python / PyTorch / FastAPI / Docker • Pre-Sales to Post-Sales Technical Delivery

PROFESSIONAL EXPERIENCE

Progress Solutions Inc. *Jul 2025 - Present*

Jr. AI/ML Engineer - Healthcare Technology, AI Consulting, SaaS Healthcare (USA)

- Led **end-to-end deployment of LLM systems** for healthcare enterprise: scoped clinical workflow needs, designed RAG architecture, built evaluations, deployed to production.
- Built **RAG systems** for clinical knowledge retrieval with data provenance and audit trails; improved precision **~35%**, grounding alignment exceeded **90%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for clinical risk stratification (patient no-show, care engagement); improved recall **15-20%** on high-risk categories; precision held **>90%**.
- Maintained **HIPAA-conscious data governance**: de-identification, data lineage documentation, audit trails, on-call readiness for deployed systems.
- Translated ambiguous healthcare workflow needs into hypothesis-driven system requirements and production solutions. Packaged as FastAPI/Flask APIs with Docker; reduced post-deploy defects **~30%**.

Suvidha Foundation *Jan 2022 - Mar 2022*

Machine Learning Engineer

- Transformer-based domain adaptation for video analytics; RAG document Q&A with >85% accuracy.

Energy Solutions International (Emerson) *Jun 2022 - Apr 2023*

Database & DevOps Performance Engineer (Intern)

- Enterprise SaaS system optimization, CI/CD automation, Grafana observability in compliance-sensitive (oil & gas) environment.

KEY PROJECTS

Agentic Claims Processing - Production Life Sciences AI Analog

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate Detection → Risk Scoring) with schema-validated agent contracts. **RAG-grounded regulatory validation** against vector-indexed policy documents. Explainable reasoning traces with data provenance - directly analogous to OpenAI FDE deployments in clinical submissions and drug discovery workflows.

IoT + ML Smart Building - Time-Series Forecasting

Sensor data pipeline with walk-forward validation; Random Forest, SVR, Linear Regression comparison; lag-window feature engineering. **~20-30% lower MAE** vs linear baseline. Demonstrates ML hypothesis testing and experimental rigor.

Driver Drowsiness Detection - YOLOv8 Real-Time System

Unified YOLOv8 detection pipeline; ~30% latency reduction; sliding-window confidence aggregation for false positive suppression.

EDUCATION

Master of Science, Computer Science - *Kent State University, Ohio, USA* *Aug 2023 - May 2025*

SKILLS

Languages: Python, TypeScript, SQL

LLM & AI: RAG, agentic workflows, LangChain, LlamaIndex, prompt engineering, evaluation pipelines, data provenance

ML: PyTorch, scikit-learn, XGBoost, HuggingFace, time-series forecasting

Infra: Docker, FastAPI, Flask, Jenkins, Grafana, REST APIs

Work Authorization: F-1 OPT (US-authorized now); open to work authorization discussions for EU roles